

HENRY DOWD

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PROFILE

BSc in Pure Mathematics & Computer Science, Maynooth University, completed May 2026, with a year abroad at the University of Ottawa. Run a two-node k3s cluster on a single Proxmox host, administering the stack from physical hardware and virtualisation up to GitOps-managed Kubernetes, with CI/CD, monitoring, and alerting built in. Comfortable with Linux administration, networking, and Python scripting, and troubleshoot my own infrastructure when it breaks. Targeting roles in platform engineering, site reliability, DevOps, and low-level infrastructure.

TECHNICAL SKILLS

Linux & Virtualisation Arch Linux, Proxmox VE, LXC, ZFS, systemd, shell scripting, SSH key management
Networking TCP/IP, DNS (split-horizon), DHCP, WireGuard, Cloudflare Tunnels, reverse proxies / ingress, SSL/TLS
Containers & CI/CD Kubernetes (k3s), Docker, Helm, ArgoCD (GitOps), GitHub Actions, MetalLB, Traefik, Longhorn
Observability & Backup VictoriaMetrics, Grafana, Alertmanager, restic, Backblaze B2
Scripting & Tools Python, Bash, Java, C, SQL; Git, Gitea, cert-manager (Let's Encrypt), Technitium DNS, $\LaTeX$

PROJECTS

Self-Hosted Infrastructure & Home Lab · Personal

2024 – Present

- Two-node k3s cluster on single Proxmox host, GitOps via ArgoCD app-of-apps pattern, git repo as single source of truth.
- Platform assembled from components: MetalLB load balancing, Traefik ingress, Longhorn persistent block storage (replica=1, deliberate single-worker tradeoff), Sealed Secrets for encrypted credentials in git.
- CI/CD end-to-end: GitHub Actions builds, pushes to GHCR, updates SHA-pinned manifest, ArgoCD reconciles; in-cluster runner lints manifests.
- Public ingress via Cloudflare Tunnel (no inbound ports open), split-horizon DNS via Technitium, wildcard TLS via cert-manager and Let's Encrypt DNS-01.
- Observability stack VictoriaMetrics, Grafana, Alertmanager with `absent()` guards so a dead scrape self-alerts; encrypted offsite restic backups to Backblaze B2, quarterly verified restores.
- Diagnosed intermittent e1000e (Intel I219-LM) NIC transmit hangs across ethtool offload settings, modprobe driver parameters, and GRUB `pcie_aspm=off` — layered isolation separating configuration from firmware.
- Decisions documented as ADRs, incidents as post-mortems. github.com/hpdowd/homelab

Final Year Project — Paraphrase Detection · Maynooth University

2025 – 2026

- Built a dual-branch classifier fusing BERT cosine similarity, Zhang–Shasha tree edit distance, Jaccard overlap, and length ratio; evaluated on the MSRP corpus in Python with HuggingFace Transformers, PyTorch, and scikit-learn.
- Performed error analysis comparing structural vs. semantic methods, with confusion matrices, score distributions, and parse-tree examples of failure modes. github.com/hpdowd/paraphrase_detector

PDF-Sorter — Workplace Automation Tool · Python

2025

- Regex-based tool to auto-sort and rename PDFs by filename and content metadata, with OCR fallback; deployed during HR placement at Roscommon County Council. github.com/hpdowd/PDF-Sorter

EXPERIENCE

Summer Student — Human Resources · Roscommon County Council

Jun – Sep 2025

- Data entry, records management, and financial processing across employee files and payroll-adjacent administration.
- Identified a repetitive document-sorting bottleneck and built a custom Python tool to automate the workflow.

EDUCATION & CERTIFICATIONS

BSc Computational Thinking · Maynooth University

Sep 2022 – May 2026

Pure Mathematics & Computer Science

Study Abroad · University of Ottawa

Sep 2024 – May 2025

Fortinet NSE 1–3 (2026) · Full driving licence